

# Skills Summary

## Adding and Subtracting Mixed Numbers

*Use this reference while completing your worksheet or when studying.*

**A mixed number is a whole plus a fraction**, so work the two parts in columns: wholes with wholes, fractions with fractions, then put them back together.

**Same-size pieces rule:** fractions can only be added or subtracted when the denominators match. The denominator names the size of the piece and never changes when you add or subtract.

**One whole is always the fraction with equal top and bottom** — three thirds, four fourths, eight eighths. That is what makes trading possible in both directions.

### 1. Adding with the same denominator

Add the wholes, add the fractions, and then look at the fraction part. If it is one whole or more, trade it: carry the whole across and keep what is left. Finish in simplest form, with the unit from the story.

**Example:** A hike is  $3\frac{2}{5}$  km, then  $1\frac{4}{5}$  km. How far in all?

**Step 1.** The fifths match, so the parts can be added as they stand — but two fifths plus four fifths passes one whole, so expect a trade.

**Step 2.** Wholes:  $3 + 1 = 4$ . Fifths:  $\frac{2}{5} + \frac{4}{5} = \frac{6}{5} = 1\frac{1}{5}$ , and  $4 + 1\frac{1}{5}$  is the total.  
The hike is  $5\frac{1}{5}$  km.

**Try it:** A ribbon of  $2\frac{3}{8}$  m is joined to one of  $4\frac{6}{8}$  m. How long is the joined ribbon?

$\frac{8}{1}$  check

### 2. Subtracting when you have to regroup

If the fraction you are taking away is bigger than the one you have, borrow one whole and change it into a fraction with the same denominator, then add it to the fraction you already had.

Never turn the subtraction around. Reversing the fraction column answers a different question and always gives an answer that is too big.

**Example:** A jug holds  $5\frac{1}{4}$  L. Pour out  $1\frac{3}{4}$  L. How much is left?

**Step 1.** Three quarters cannot come out of one quarter, so borrow a whole before subtracting anything.

**Step 2.**  $5\frac{1}{4} = 4\frac{5}{4}$ . Wholes:  $4 - 1 = 3$ . Quarters:  $\frac{5}{4} - \frac{3}{4} = \frac{2}{4} = \frac{1}{2}$ .  
There is  $3\frac{1}{2}$  L left.

**Try it:** A board is  $6\frac{1}{5}$  ft long. A piece of  $2\frac{3}{5}$  ft is cut off. How much board is left?

$\frac{5}{3}$  check

### 3. When the denominators are different

Rewrite both fractions with a common denominator first — the smallest number both denominators divide is easiest — and only then add or subtract in columns.

After rewriting, check the fraction column again: a subtraction may still need regrouping, and a sum may still need a trade.

**Example:** A cook mixes  $2\frac{1}{2}$  cups of stock with  $1\frac{1}{3}$  cups of cream. How much liquid is that?

**Step 1.** Halves and thirds are different-sized pieces, so nothing can be added until both are written over one denominator, and sixths is the smallest that works.

**Step 2.**  $\frac{1}{2} = \frac{3}{6}$  and  $\frac{1}{3} = \frac{2}{6}$ . Wholes:  $2 + 1 = 3$ . Sixths:  $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$ .

The mixture is  $3\frac{5}{6}$  cups.

**Try it:** A tank holds  $4\frac{2}{3}$  gal and  $1\frac{1}{4}$  gal is drained. How much is left?

check:  $\frac{21}{9}$

(!) **Watch out:** adding or subtracting straight across the denominators is the most common slip. Estimate with the nearest whole numbers before you start, and check that your answer lands near the estimate.